1-1. Maxwel eqs.

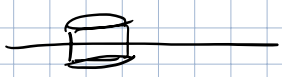
$$\nabla \cdot D = \rho \qquad \oiint D\,dS = \iiint \rho\, dV$$

$$\nabla \cdot B = 0 \qquad \oiint B\,d\Sigma = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \qquad \oint E\,dl = -\iint \frac{\partial B}{\partial t}\,d\sigma$$

$$\nabla \times H = J + \frac{\partial D}{\partial t} \qquad \oint H\,dl = \iint \left(J + \frac{\partial D}{\partial t}\right) d\sigma$$

1-2.



$$(D_1 - D_2)\,n = 0$$

1-5.

$$w = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu} B^2\right)$$

$$= \frac{\varepsilon(1+\mu)}{2}\left[A^2\cos^2(kz-\omega t) + B^2\sin^2(kz-\omega t)\right.$$

$$S = E \times H = \begin{vmatrix} x & y & z \\ E_x & E_y & E_z \\ H_x & H_y & H_z \end{vmatrix} = \sim.$$

1-6

$D = \varepsilon E$. $B = \mu H$.

$\varepsilon = \varepsilon(r)$ $\mu = \mu(r)$.

$\nabla \cdot D = \rho = 0$ $\nabla \cdot B = 0$

$\nabla \times E = -\frac{\partial B}{\partial t}$ $\nabla \times H = +\frac{\partial D}{\partial t}$.

$$\nabla \times (\nabla \times E) = \nabla(\nabla \cdot E) - \nabla^2 E.$$

$$= -\frac{\partial}{\partial t}(\nabla \times B) = -\frac{\partial}{\partial t}(\nabla \times \mu H)$$

$$= -\frac{\partial}{\partial t}\mu \nabla \times H - \frac{\partial}{\partial t}\nabla\mu \times H$$

$$= +\mu\varepsilon\omega^2 E + \frac{\partial}{\partial t}\nabla\mu \times \frac{B}{\mu}$$

$$= \mu\varepsilon\omega^2 E + \nabla \ln\mu \times \frac{\partial}{\partial t} B = \mu\varepsilon\omega^2 E - \nabla \ln\mu \times (\nabla \times E)$$

又 $\nabla \cdot D = \nabla \cdot (\varepsilon E) = (\nabla\varepsilon) \cdot E + \varepsilon \nabla \cdot E$.

$\nabla \cdot D = \rho = 0$

$$= (\nabla\varepsilon)\frac{D}{\varepsilon} + \varepsilon \nabla \cdot E = (\nabla \ln\varepsilon) D + \varepsilon \nabla \cdot E = 0$$

$$\nabla \cdot E = -\frac{1}{\varepsilon}(\nabla \ln\varepsilon) D = -(\nabla \ln\varepsilon) E$$

$$\nabla(\nabla \cdot E) = -\nabla[(\nabla \ln E) E]$$

$\therefore$ $\nabla^2 E + \nabla[E \cdot (\nabla \ln E)] + \mu\varepsilon\omega^2 E + \nabla \ln\mu \times (\nabla \times E) = 0$